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Ratio & Proportion

$\frac{a}{b}$ → numerator, antecedent, first term
→ denominator, consequent, second term
↳ a to b / or / a : b

→ If we multiply or ÷ the numⁿ or denoⁿ of ratio by same no :- ration has no change

$$\frac{a}{b} = \frac{a \times 5}{b \times 5} = \frac{a}{b}$$

Q: If $p : q = r : s = t : u = 2 : 3$ then what is value of $(p + r + t) / (q + s + u)$?

Ans

$$\frac{p + r + t}{q + s + u} = \frac{2}{3}$$

or

$p = \frac{2q}{3}$ $\quad r = \frac{2s}{3}$ $\quad t = \frac{2}{3}u$

$$\frac{\frac{2q}{3} + \frac{2}{3}s + \frac{2}{3}u}{q + s + u} = \frac{2}{3} \frac{\cancel{(q + s + u)}}{\cancel{q + s + u}} = \frac{2}{3}$$

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Properties

* $\frac{k_1}{k_2} = \frac{k_3}{k_4} = \frac{k_5}{k_6} = \frac{k_7}{k_8}$

* $\frac{k_1 + k_2 + k_3}{k_4 + k_5 + k_6} = \frac{k_7}{k_8}$

* $\frac{k_1 + k_3 + k_4}{k_2 + k_4 + k_6} = \frac{k_5}{k_6}$

* $\frac{k_1 + k_3}{k_2 + k_4} = \frac{k_5}{k_6}$

Q: If $a/b = c/d = 2/3$ then what is value of $\frac{ab+cd}{b^2+d^2}$?

A

$$\frac{a}{b} = \frac{c}{d} = \frac{2}{3}$$

$$\frac{a+c}{b+d} = \frac{2}{3} \rightsquigarrow \frac{ab+cd}{b^2+d^2} = \frac{2}{3}$$

- $\frac{a k_1 + b k_3 + c k_5}{a k_2 + b k_4 + c k_6} = \frac{k_7}{k_8}$

Q: If $p/q = r/s = t/u = 2/3$ then what is value of $(mp + nr + ot) : (mq + ns + ou)$?

$\frac{2}{3}$ Ans

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Types of Ratio:-

Let ratio be expressed as $a:b$.

Type 1 Duplicate Ratio = a^2/b^2

Type 2 Sub-Duplicate Ratio = $\sqrt{a}/\sqrt{b}$

Type 3 Triplicate ratio = a^3/b^3

Type 4 Sub-Triplicate Ratio = $\sqrt[3]{a} : \sqrt[3]{b}$

Type 5 Inverse / Reciprocal Ratio = $1/a : 1/b$

$a:b:c \rightarrow \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$

(Take LCM)

$bc : ac : ab$

Compound / Compounded / Mixed Ratio:-

C.R of $m:p$ & $p:q$ is $m \times p : p \times q$

$m:p$ & $p:q$ & $r:s \rightarrow m \times p \times r : p \times q \times s$

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Q: find C.R of 2/3, 7/9 & 12/35

$$\frac{2\times7\times12}{3\times9\times35}$$

Q: If a:b = 2:3, b:c = 5:6 find a:b:c?

a	b	c
2	3	3
5	5	6
10	15	18

a:c = 10:18

Q: If a:b = 3:4, b:c = 5:6 & c:d = 2/3 find a:b:c:d?

A

a	b	c	d
3	4	4	4
5	5	6	6
2	2	2	3
30	40	48	72

Q: If ratio of x to y = 6/5 & ratio of z to y = 3/4, ratio of x to z is?

A

x	y	z
6	5	5
4	4	3
24	20	15

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